

Program Risk Opportunity and Issue Register

Lunar Surface Infrastructure Program

EXPOSURE OWNERSHIP RESPONSE DECISIONS

Owner	Program Director	Status	Draft for review
Version	0.1	Control account	WBS 1.1.5
Coverage	Programwide	Date	23 September 2026
Basis	Charter v1.1; WBS v0.2; robotic control artifacts v0.1	Next review	At initial program risk review

Purpose: Establish the initial programwide baseline for identifying, scoring, owning, responding to, monitoring, and escalating risks, opportunities, and current issues across the Lunar Surface Infrastructure Program.

Management conclusion: The most significant preliminary exposures concern site knowledge, power and thermal continuity, transportation sequencing, robotic reliability, cross-system interfaces, planning maturity, and crew survival readiness. The register provides a disciplined control structure; accountable owners must validate every score and convert responses into scheduled work.

1 Executive Control Summary

Decision requested: Approve this register as the working baseline for initial owner validation and response planning. Approval establishes management discipline and review cadence; it does not approve mission design, funding, contractual commitments, or crewed operations.

Measure	Current count	Director interpretation
Open risks	18	Initial planning exposure requiring owner validation and response planning.
High risks	10	Require weekly review until exposure reduces or the decision authority accepts the risk.
Medium risks	8	Require active owner actions and monthly program-level review.
Low risks	0	Monitor for changes; no risks currently score Low before planned responses.
Open issues	3	Current planning gaps already present and requiring closure actions.
Proposed opportunities	7	Candidate investments or design choices that may improve readiness and resilience.

Initial director priorities

- Validate the ten High risks with the accountable owners and confirm whether the proposed response changes probability, impact, or both.
- Close the three current planning issues before using the program model to support schedule or readiness commitments.
- Protect crew survival and safe departure as non-negotiable readiness conditions; unresolved critical evidence remains a no-go condition.
- Integrate risk responses with the WBS, responsibility assignments, milestone network, future integrated master schedule, and simulator roadmap.
- Evaluate the seven opportunities through defined decision points before adding scope or consuming resources.

Program control principle

Risk management must influence decisions, scope, schedule, verification, and reserves. A register that records exposure without changing work or decision criteria provides limited control value.

2 Scoring Method and Governance

Probability describes the chance that the event occurs within the planning horizon. Impact represents the most credible consequence to crew safety, mission capability, schedule, cost, technical performance, or program reputation. Owners must document the basis for both values when they validate the register.

Probability and impact matrix

Score equals probability multiplied by impact. Initial scores represent exposure before the planned response. Residual targets are management objectives and require validation after response actions are defined and implemented.

Probability / Impact	1 Minimal	2 Minor	3 Moderate	4 Major	5 Critical
5 Almost certain	5 Low	10 Medium	15 High	20 High	25 High
4 Likely	4 Low	8 Medium	12 Medium	16 High R-01, R-06, R-07, R-12, R-14	20 High R-02, R-03
3 Possible	3 Low	6 Medium	9 Medium R-15	12 Medium R-05, R-08, R-11, R-13, R-17	15 High R-04, R-09, R-10
2 Unlikely	2 Low	4 Low	6 Medium	8 Medium	10 Medium R-16, R-18
1 Rare	1 Low	2 Low	3 Low	4 Low	5 Low

Thresholds and management action

Rating	Score	Required management action
High	15-25	Review weekly; establish funded or scheduled response actions; escalate unresolved exposure to the Program Director or decision authority.
Medium	6-14	Review monthly; maintain owner actions and triggers; escalate when the trend worsens or a decision gate approaches.
Low	1-5	Monitor quarterly or at relevant decision gates; retain the basis for acceptance or watch-list status.

Governance rules

- The Program Director owns the program-level process; functional owners own evidence, actions, triggers, and forecasts for assigned entries.
- An issue requires immediate containment and closure planning because the condition already exists.
- An opportunity requires a benefit hypothesis, accountable owner, decision point, and resource decision before incorporation into the baseline.
- A risk closes only when the event can no longer occur, the exposure becomes an accepted residual risk, or the condition converts to an issue.
- Changes to scope, schedule, interfaces, requirements, or readiness criteria follow configuration and baseline change control.

3 Risk Identification and Initial Scoring

The register uses cause-event-effect statements. Scores remain preliminary until accountable owners confirm probability, impact, assumptions, and response feasibility.

ID	WBS	Risk	Cause event and effect statement	Owner	P	I	Score	Rating
R-01	1.1.3	Uneven planning maturity	If the habitat, transportation, data-center, and shared-infrastructure branches remain less decomposed than robotic deployment, integration decisions may rely on incomplete scope and hidden dependencies.	PMO	4	4	16	High
R-02	1.3.2	Site evidence uncertainty	If terrain, subsurface, illumination, thermal, radiation, dust, and hazard data remain incomplete, the program may select a site that cannot support safe installation or sustained operations.	SCI	4	5	20	High
R-03	1.5.2 / 1.6.1	Power continuity shortfall	If generation, storage, distribution, or load-priority margins are insufficient, critical habitat and computing loads may experience extended power deficits.	SIO	4	5	20	High

ID	WBS	Risk	Cause event and effect statement	Owner	P	I	Score	Rating
R-04	1.5.3 / 1.6.3	Thermal rejection constraint	If heat-rejection capability cannot accommodate operating loads and extreme thermal conditions, computing capacity may require prolonged derating or shutdown.	DCO	3	5	15	High
R-05	1.3.3 / 1.6.2	Communications or navigation outage	If surface communications, positioning, or timing services lose coverage, robotic work and remote recovery may stop or continue with unacceptable uncertainty.	NET	3	4	12	Medium
R-06	1.2	Transportation and cargo dependency	If launch, trans-lunar delivery, landing, or cargo sequencing changes, critical equipment may arrive late, damaged, or in an unusable deployment order.	LOG	4	4	16	High
R-07	1.3.1 / 1.3.6	Robotic degradation and mobility loss	If dust, terrain, temperature, wear, or hardware faults degrade the robotic fleet, the program may not complete survey, preparation, assembly, or repair work before crew arrival.	ROB	4	4	16	High

ID	WBS	Risk	Cause event and effect statement	Owner	P	I	Score	Rating
R-08	1.3.1 / 1.3.6	Autonomy and recovery limitation	If autonomy, fault isolation, or remote recovery cannot manage delayed or interrupted control, robotic systems may enter unsafe states or require unavailable intervention.	MOPS	3	4	12	Medium
R-09	1.4.1-1.4.7	Habitat readiness misalignment	If habitat commissioning, life support, provisions, emergency systems, or training lag the crewed mission decision, crew arrival may precede verified survival capability.	HAB	3	5	15	High
R-10	1.4.4 / 1.6.5	Radiation event exposure	If a solar particle event or elevated radiation occurs without adequate detection, shielding, or shelter capacity, crew health and electronic systems may exceed approved exposure limits.	SAFE	3	5	15	High
R-11	1.4 / 1.5 / 1.6	Micrometeoroid damage	If an impact breaches or disables a critical asset, the program may lose pressure integrity, power, communications, thermal control, or computing capacity.	SAFE	3	4	12	Medium

ID	WBS	Risk	Cause event and effect statement	Owner	P	I	Score	Rating
R-12	1.1.4	Cross-system interface mismatch	If mechanical, electrical, software, thermal, data, or operational interfaces mature independently, delivered systems may not connect, share services, or recover together.	SE	4	4	16	High
R-13	1.3.6 / 1.6.4	Insufficient repair and spares coverage	If spares, tools, maintenance data, and remote repair capability do not cover credible failures, a recoverable fault may become a long-duration outage.	LOG	3	4	12	Medium
R-14	1.1.3	Unvalidated integrated schedule	If durations, calendars, resources, and branch logic are not loaded and validated, the program may misidentify the controlling path and readiness forecast.	PMO	4	4	16	High
R-15	1.5.7	Simulator interpretation risk	If users treat the educational simulator as a mission-grade design tool, simplified assumptions may drive unsupported conclusions about power, thermal, or availability performance.	DCO	3	3	9	Medium

ID	WBS	Risk	Cause event and effect statement	Owner	P	I	Score	Rating
R-16	1.5.5 / 1.6.2	Cyber or command-path compromise	If identity, access, command authority, network segmentation, or data protection controls fail, an unauthorized or corrupted command could disrupt critical systems.	NET	2	5	10	Medium
R-17	1.3.4 / 1.3.5	Concurrent surface-work interference	If survey, grading, networking, utilities, and assembly overlap without controlled work zones, one activity may damage assets or create unsafe conditions for another.	SUR	3	4	12	Medium
R-18	1.1.6 / 1.4	Premature crew mission commitment	If external pressure drives a crewed mission commitment before integrated readiness evidence is accepted, the program may absorb unacceptable survival and recovery risk.	PD	2	5	10	Medium

4 Risk Response Planning

The response register defines the management approach, planned action, early warning trigger, and residual target. Owners must convert each planned response into controlled activities with dates, resources, and closure evidence.

ID	Owner	Strategy	Planned response	Trigger or early warning	Residual target
R-01	PMO	Mitigate	Complete selective Level 4 decomposition, responsibility assignment, and milestone logic for branches that control readiness or common services.	A decision gate lacks an accountable work package, predecessor, or measurable exit criterion.	P2 x I4 = 8
R-02	SCI	Mitigate	Define site-selection criteria, survey coverage, data-quality thresholds, alternate sites, and a formal site-suitability review.	Survey data fail coverage, confidence, or hazard-tolerance criteria at the planned site decision.	P2 x I5 = 10
R-03	SIO	Mitigate	Establish critical-load tiers, energy-margin requirements, degraded operating modes, emergency power, and recovery criteria using integrated power simulations.	Forecast state of charge or generation margin crosses the approved critical-load threshold.	P2 x I5 = 10
R-04	DCO	Mitigate	Allocate thermal margins, model seasonal and degraded cases, define load-shedding logic, and verify redundant heat-transport paths.	Predicted equipment temperature or radiator margin approaches an approved operating limit.	P2 x I4 = 8
R-05	NET	Mitigate	Use phased network deployment, coverage verification, redundant paths, local autonomy, and defined safe states for loss of link or position.	Coverage, latency, time synchronization, or position error exceeds the approved operating limit.	P2 x I4 = 8
R-06	LOG	Mitigate	Create manifest priorities, delivery sequence constraints, substitute configurations, protected margins, and contingency plans for partial or delayed delivery.	A delivery slips beyond its required surface need date or a payload fails integration or landing acceptance.	P3 x I3 = 9
R-07	ROB	Mitigate	Provide mobility margin, fleet redundancy, dust controls, repair tooling, spares, health monitoring, and recovery demonstrations under representative conditions.	Fleet availability, mobility margin, or task completion rate falls below the approved threshold.	P2 x I4 = 8

ID	Owner	Strategy	Planned response	Trigger or early warning	Residual target
R-08	MOPS	Mitigate	Verify fault-management logic, command authority, safe modes, diagnostic coverage, rollback criteria, and operator recovery procedures.	A fault cannot be isolated or recovered within the approved time without direct local intervention.	P2 x I4 = 8
R-09	HAB	Avoid	Require objective crew-readiness evidence, independent verification, margin for 60 days of critical provisions, and a no-go decision for unresolved survival-critical conditions.	Any survival-critical verification remains incomplete at the crew mission readiness review.	P1 x I5 = 5
R-10	SAFE	Mitigate	Define monitoring and warning thresholds, storm-shelter capacity, radiation-hardened critical electronics, exposure procedures, and recovery inspections.	Forecast or measured radiation exceeds the action threshold or monitoring becomes unavailable.	P2 x I5 = 10
R-11	SAFE	Mitigate	Use shielding, physical separation, leak and damage detection, isolation zones, repair kits, redundancy, and recovery exercises.	Impact detection, unexplained pressure loss, or sudden component degradation indicates possible damage.	P2 x I4 = 8
R-12	SE	Mitigate	Control interface requirements, maintain interface owners, conduct cross-branch design reviews, and verify end-to-end interfaces before surface delivery.	An interface requirement remains unowned, changes without approval, or fails integrated verification.	P2 x I4 = 8
R-13	LOG	Mitigate	Use failure-mode analysis to size spares, standardize replaceable units, verify repair access, and prioritize replenishment by mission consequence.	A credible single-point failure has no repair method, spare, or approved degraded operating mode.	P2 x I3 = 6
R-14	PMO	Mitigate	Build the integrated master schedule from approved work packages, validate logic and resources, assess schedule risk, and control baseline changes.	A decision or commitment depends on a date that lacks validated logic, resource loading, or schedule margin.	P2 x I4 = 8
R-15	DCO	Mitigate	Maintain model disclosures, document assumptions, label scenario outputs, version model changes, and separate teaching results from engineering decisions.	A published conclusion omits model limitations or uses outputs to claim mission readiness.	P1 x I3 = 3

ID	Owner	Strategy	Planned response	Trigger or early warning	Residual target
R-16	NET	Mitigate	Define trusted command paths, least-privilege access, authentication, segmentation, monitoring, safe command rejection, and incident recovery.	Unauthorized access, anomalous command activity, integrity failure, or loss of trusted time and identity services.	P1 x I5 = 5
R-17	SUR	Mitigate	Use work-zone control, route and exclusion maps, configuration holds, sequencing rules, and positive release of areas between teams.	Two activities require the same route, work area, service, or configuration at the same time.	P2 x I3 = 6
R-18	PD	Avoid	Maintain independent readiness criteria, explicit no-go conditions, residual-risk acceptance authority, and documented decision records.	A crew mission date is proposed while critical readiness evidence, contingencies, or safe-departure capability remain open.	P1 x I5 = 5

5 Opportunity Register

Opportunity scores use likelihood multiplied by potential benefit. High scores identify candidates for near-term evaluation; they do not authorize funding or scope changes.

ID	WBS	Opportunity	Potential program benefit	Owner	L	B	Score	Next action
O-01	1.5.6 / 1.6.7	Accelerated hazard simulation	Use compressed solar, dust, radiation, impact, power, and thermal events to test fault isolation, load shedding, recovery, and design trades before field execution.	DCO	5	5	25	Develop the event engine and connect failures to measurable recovery KPIs. Decision point: Simulator hazard module review
O-02	1.3	Robotic precursor operations	Deploy robotic systems early enough to characterize the site, prepare infrastructure, resolve hazards, and reduce work assigned to the crew.	ROB	4	5	20	Prioritize precursor tasks by crew-risk reduction and readiness value. Decision point: Robotic mission concept review
O-03	1.1.4	Standardized modular interfaces	Adopt common mechanical, power, data, thermal, and service interfaces to reduce integration effort and support replacement or expansion.	SE	4	4	16	Establish interface standards and test representative cross-system connections. Decision point: Interface control baseline
O-04	1.3.2	High-value polar site selection	Use evidence-based site selection to improve illumination, thermal conditions, communications geometry, and access to scientifically or operationally useful areas.	SCI	3	5	15	Compare candidate sites using weighted readiness criteria and uncertainty ranges. Decision point: Lunar site characterization review

ID	WBS	Opportunity	Potential program benefit	Owner	L	B	Score	Next action
O-05	1.3.6 / 1.5.6	Predictive maintenance	Use health data and degradation trends to schedule maintenance before failures interrupt critical services.	MOPS	4	4	16	Define health indicators, alert thresholds, remaining-life logic, and maintenance decision rules. Decision point: Remote maintenance readiness review
O-06	1.6.1 / 1.6.3	Shared power and thermal capacity	Coordinate shared services so unused margin can support the most critical operating loads during degraded conditions.	SIO	4	5	20	Define allocation rules, isolation boundaries, service priorities, and cross-system recovery modes. Decision point: Shared infrastructure architecture review
O-07	1.1.6 / 1.6.7	Phased readiness demonstrations	Use incremental demonstrations to expose integration gaps before the final uncrewed and crewed readiness decisions.	MA	4	5	20	Build an evidence ladder from component tests through integrated degraded-mode and recovery demonstrations. Decision point: Verification strategy approval

6 Issue Register

Issues represent current conditions, not uncertain future events. Each issue remains open until objective closure evidence is accepted by the named authority.

ID	Current condition	Program effect	Owner	Immediate action	Closure evidence	Status
I-01	The current simulator does not yet model accelerated natural hazards, hardware degradation, cascading failures, or recovery actions.	Limits resilience analysis and prevents direct comparison of downtime-reduction strategies.	DCO	Define the hazard-event architecture, failure mapping, and recovery controls for the next model release.	Demonstrated scenario set with traceable events, failures, operator actions, and recovery KPIs.	Open
I-02	The preliminary integrated milestone network contains dependency logic but no durations, dates, calendars, resources, float, or confirmed critical path.	The program cannot yet produce a validated readiness forecast or quantify schedule margin.	PMO	Collect owner estimates, load resources and calendars, validate logic, and perform schedule-risk analysis.	Approved integrated master schedule with a documented controlling path and schedule margins.	Open
I-03	Robotic deployment has Level 4 work packages, responsibility assignments, and milestone logic; the other Level 2 branches do not yet have equivalent control detail.	Cross-branch interfaces and readiness dependencies may remain hidden until late planning.	PD	Prioritize selective decomposition for transportation, habitat, data center, and shared infrastructure based on risk and gate needs.	Approved control packages for all elements that drive integrated readiness decisions.	Open

7 Review Cadence and Next Decisions

Operating cadence

Forum	Frequency or timing	Primary purpose	Required output
Program risk review	Weekly while High risks remain open	Validate trends, action status, triggers, and escalation needs.	Updated register and decisions
Integrated baseline review	Monthly	Assess risk effects on scope, schedule, resources, interfaces, and reserves.	Baseline change actions
Technical and readiness reviews	At defined gates	Confirm entry criteria, evidence, residual exposure, and no-go conditions.	Gate disposition and accepted residual risks
Simulator model review	Each controlled release	Confirm assumptions, fidelity changes, scenario coverage, and model limitations.	Release record and model baseline
Executive show-and-tell	At selected maturity points	Communicate progress, invite informed challenge, and identify collaboration opportunities.	Feedback and follow-up actions

Initial decisions for authorization

- Approve the 5 by 5 scoring method and escalation thresholds as the working program standard.
- Confirm accountable owners and require validation of initial scores, triggers, responses, and residual targets.
- Authorize selective Level 4 decomposition for program elements that control integrated readiness or High-risk responses.
- Authorize development of the integrated master schedule and schedule-risk analysis after owner logic and duration estimates mature.
- Authorize the simulator hazard module as an educational test environment for resilience, downtime, and recovery strategies.

Configuration and reporting

The PMO will maintain a controlled master register, preserve revision history, and link material entries to WBS elements, work packages, milestones, decision records, verification evidence, and simulator releases. Public-facing material will summarize approved educational content without disclosing unsupported mission conclusions.

Version record

Version	Date	Change	Disposition
0.1	23 September 2026	Initial programwide risk, opportunity, and issue baseline.	Draft for review

Educational planning model. This artifact supports learning and program-management practice; it does not constitute mission authorization, engineering certification, or operational approval.